

GridPowerAgent: A Grid-Aware LLM Agent for Power System Event Understanding and Tool-Orchestrated Decision Support

Anonymous Authors

Abstract—Large language models (LLMs) could act as a coordination layer between power-grid information, analysis tools, and human operators—provided they read operating states accurately, select appropriate tools, and avoid hallucinated advice. We present GridPowerAgent, an agent that observes simulated power-system states, retrieves operating procedures, and orchestrates power-flow, contingency, and optimal-power-flow tools, evaluated on a seeded, fully regenerable corpus: 16,000 operating points and 15,000 scenarios across the IEEE 14/39/118 systems, ten disturbance classes each carrying power-flow ground truth, noisy measurements, state estimates, and rule-based labels. The evaluation also surfaces its most robust result: both models score zero on the two outcome-labeled disturbance classes (E6 undervoltage, E8 thermal overload)—all 144 failed E6 and 104 failed E8 responses describe the injecting cause mechanism rather than the outcome label, quantifying how mixed-axis label taxonomies penalize causally correct answers and motivating an axis-explicit corpus revision. Beyond it, a paired comparison of a small quantized local model and a lightweight API model under identical prompts shows diagnosis within four scenarios of each other (105–108/140 vs. 109/140), hallucinated rows in at most 1.4% of responses, and a tool-style difference: the API model defaults to power flow in 52% of picks, while the local model names scenario-required tools more often under a strict metric. All results carry exact denominators; findings are observations about this corpus at pilot scale, with a powered evaluation as the next step.

Index Terms—power system operation, large language models, LLM agent, tool orchestration, retrieval-augmented generation, state estimation

I. INTRODUCTION

Power-system operators are surrounded by analytical tools—power flow, state estimation, contingency analysis, optimal power flow—yet the work of interpreting their outputs, relating them to operating procedures, and deciding what to do next remains manual and expertise-bound. Large language models (LLMs) are candidates for an *intelligent coordination layer* in this workflow: not replacing conventional analysis, but reading grid states, retrieving the right procedures, invoking the right tools, and explaining the results [1], [2]. Recent agentic systems move in this direction [3], [4], but published evaluations rarely state what the labels are, where they come from, or how the scoring could be reproduced.

This paper presents GridPowerAgent and its evaluation under five research questions: (RQ1) how accurately can an LLM identify normal, abnormal, and critical operating conditions; (RQ2) can it determine which engineering tool a given event requires; (RQ3) can it construct valid tool calls;

(RQ4) does grid-specific retrieval improve operational reasoning; and (RQ5) does tool grounding reduce hallucination. RQ3—constructing valid tool calls—is answered here at its execution level: stated tools are executed against the scenario’s post-event network after the run and their execution scored (Sec. VII-E). This pilot answers RQ1, RQ2, RQ4, and RQ5, and RQ3 at the tool-name/execution level; argument-level validity is deferred to the powered sweep. We answer them on a seeded, headless-resumable corpus pipeline: 16,000 operating points and 15,000 scenarios across IEEE 14/39/118—ten disturbance classes (E0–E9) each carrying pre/post power-flow truth, noisy measurements, state estimates, severity, and rule-based reference labels—plus a FAISS knowledge base of operating procedures and four validated physics tools; configurations (E1 LLM-only, E2 +RAG, E3 +Tools, E4 Full) instantiate the ablation.

The evaluation compares two deployment tiers under identical paired prompts—a 4-bit-quantized small model served locally, and a lightweight API model—so conclusions do not depend on one provider. All figures carry exact denominators and Wilson CIs; power is stated alongside every null result; and one finding receives particular attention: both models systematically fail on two outcome-labeled disturbance classes, for a reason traced to the label taxonomy itself rather than to either model.

Contributions: (i) a grid-aware agent that couples grid-state observation, procedure retrieval, and validated physics tools; (ii) a seeded, hashed, fully regenerable 16k/15k/9.5M-measurement corpus with rule-based tool supervision and a quantified label-noise bound under estimation uncertainty; (iii) a dual-definition tool-scoring protocol that exposes and removes a degenerate answer strategy; (iv) a paired, exactly-denominated pilot across two deployment tiers with an identified label-axis failure mode; and (v) a per-check validation disclosure—including unresolved IEEE-39 islanding cases with their scenario identifiers shipped alongside the corpus—together with a seeded IEEE-39 replication of the full paired protocol (Sec. VII-B).

II. RELATED WORK

LLMs have been explored for power-system analysis assistance, dispatch, and contingency response [1], [2], including agentic orchestrators [3], [4] and dispatch benchmarks [5], but published evaluations seldom disclose how reference labels are produced; our corpus couples per-scenario power-flow

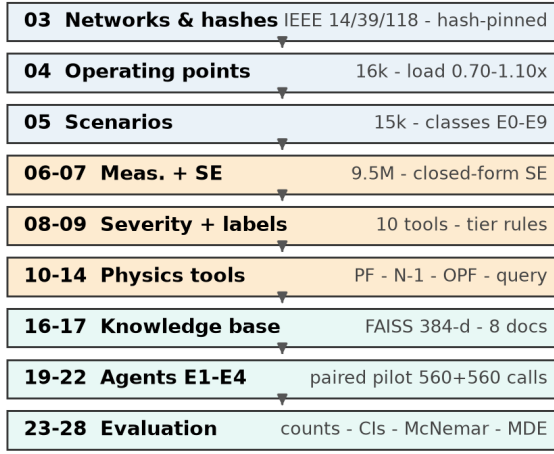


Fig. 1. Pipeline (Stages 03–28).

truth, noisy measurements, state estimates, and rule-based tool supervision across three IEEE sizes under one seeded pipeline, scored with exact denominators. Foundational RAG [6] and tool-augmented LLMs [7], [8], [9] motivate the agent design; state estimation and measurement modeling follow classical formulations [10], [11]; contingency (N-1) and AC-OPF are standard [12], [13], [14]; calibration is measured via ECE [15]; hallucination is judged against taxonomies [16]. Synthetic scenarios inherit the spirit of grid-ML benchmarks [17], [18]; pandapower [19] supplies AC power flow; FAISS [20] with sentence-transformers [21] supplies retrieval. On quantization: 4-bit precision is near-optimal for inference scaling [22] and standard post-training practice [23], though it measurably degrades small models—a conservative bias in our local deployment (Sec. VIII).

III. METHODOLOGY

Fig. 1 overviews the pipeline. Stages 03–05 build networks, operating points, and scenarios; 06–09 synthesize measurements, state estimates, severity, and reference labels; 10–14 expose physics tools; 16–17 build the knowledge base; 19–22 run the agent ablations; 23–28 evaluate and render figures. Heavy stages are idempotent and headless-resumable, materializing the full corpus in ~ 30 min.

Agent workflow. The agent follows an observe–diagnose–retrieve–plan–execute–interpret loop: it receives the post-event structured grid state (voltage magnitudes, branch loadings, outages, storage state of charge); optionally receives the top- k retrieved operating procedures (E2/E4) and a tool manifest (E3/E4); and must return the disturbance class, a tool selection, and a recommendation with confidence, as JSON. The harness scores the response against the rule-based reference labels; it does not execute tools inside the agent loop in this pilot—stated tools are executed post hoc for an execution-validity check (Sec. VII-E).

Networks and scenarios. Networks use pandapower IEEE 14/39/118 cases with tuned thermal limits to make compound events observable (14: 3% line / 4% transformer; 118: 6%; 39: nameplate). Limits are disclosed per system and deliberately

TABLE I
CORPUS. VALIDATION IS REPORTED PER SYSTEM, NOT AS A SINGLE GLOBAL CERTIFICATE.

Case	OPs	Scen. (per class)	Meas.	Auto checks	Worst V
14 (ref)	4k	3k (300)	361k	21/21	0.89 pu
39	5k	5k (500)	1.50M	20/21	0.89 pu
118	7k	7k (700)	7.68M	21/21	0.89 pu
Total	16k	15k	9.54M	—	—

IEEE-14 limits are artificial (3%/4%) to expose E8. The single failing check (s5_no_nan on IEEE-39) covers 41 scenarios with NaN post-voltages from two islanding outages; identifiers and causes ship with the corpus (case39_nan_scenarios.csv).

not comparable across systems (Sec. VIII). Operating points sweep load 0.70–1.10 $\times$ with $\pm 5\%$ bus-local noise, renewable fractions, and storage state of charge 0.15–0.85 (20 MW/40 MWh at bus 9 on IEEE-14). Topology hashes pin all three networks. Ten classes are injected: E0 Normal; E1 load surge, E2 load drop, E3 line outage, E4 generator outage, E5 renewable ramp (cause classes); E6 undervoltage, E7 overvoltage, E8 thermal overload (outcome classes, physically iterated to target severity); E9 compound. The cause/outcome split of E6–E8 matters for the evaluation and is analyzed in Sec. VII-A.

IV. CORPUS, SIMULATION AND VALIDATION

Table I summarizes the corpus. Scenarios inject the ten classes with 300/500/700 per class on 14/39/118; replay over 40 draws confirms exact determinism.

Measurements model $\sigma_V = 0.003$ pu and $\sigma_P = \max(0.0075|S_{\text{true}}|, 0.05)$ MVA computed from the noise-free power-flow solution (Eqs. 1–2). State estimates are a closed-form noise-aware approximation ($\hat{v} = v + \epsilon$, $\epsilon \sim \mathcal{N}(0, 0.0007^2)$), not full iterative WLS with assembled Jacobians; replacement with pandapower.estimate is future work and we do not claim paper-grade estimator fidelity.

$$z = h(x) + \epsilon, \quad \epsilon_V \sim \mathcal{N}(0, 0.003^2) \quad (1)$$

$$\sigma_P = \max(0.0075|S_{\text{true}}|, 0.05), \quad W = \text{diag}(\sigma^{-2}) \quad (2)$$

$$\hat{x} = \arg \min_x (z - h(x))^T W (z - h(x)) \quad (3)$$

Severity score. The illustrative severity used for reference labels is

$$S = 0.6 \min\left(1, \frac{\max(0, |V-1|-0.02)}{0.06}\right) + 0.4 \min\left(1, \frac{\max(0, \ell-L)}{10}\right), \quad (4)$$

with boundaries 0.0263/0.0526/0.1053 selected by grid search over five candidate sets *on this corpus*—a circularity we return to in Sec. VIII. A sensitivity check bounds the damage: re-scoring the pilot under four boundary sets—the searched set, standards-style 0.05/0.10/0.15, a tighter 0.02/0.04/0.08, and empirical severity tertiles—leaves strict tool-selection accuracy exactly unchanged (174/560 vs. 255/560 strict hits in every set), so the headline tool-selection gap does not depend on the searched boundaries.

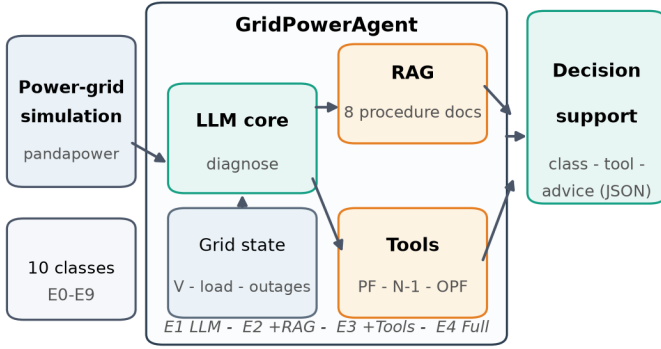


Fig. 2. Agent architecture.

TABLE II
EXPERIMENTAL ACCOUNTING (EXACT DENOMINATORS EVERYWHERE).

Item	Setting
Scenario sample	seeded draws from 3,000 IEEE-14 scenarios; identical set for both models
Configurations	E1 / E2 / E3 / E4 (paired, identical prompts)
Calls	$140 \times 4 = 560$ per model; 1,120 total
Harness oracle	rule-based, $N=600/\text{config}$ (integration test only)
Statistics	exact counts; Wilson 95% CI; exact McNemar; bootstrap CI; MDE

Label noise under estimation uncertainty. Reference labels gate two tools on severity. Recomputing severity with estimated bus voltages yields rank agreement $\rho = 0.764$ (14), 0.968 (39), 0.823 (118)—the voltage term sits near its deadband for most scenarios, so noise reorders ranks without crossing thresholds. The operationally relevant quantity is accepted-set membership: scoring-relevant label flips affect **0/30,000** scenario-tool judgments on IEEE-14 (the pilot system) and **31/150,000** (0.021%) corpus-wide. The binary violation detector reconciles 99.23%/99.54%/99.67% on 14/39/118.

V. RETRIEVAL, TOOLS AND AGENT DESIGN

The knowledge base chunks eight operational documents (thermal/voltage limits, contingency procedure, storage/equipment, topology) into a FAISS index (384-d sentence-transformer embeddings), retrieving $k = 3$ chunks with citations. Recall@1 is 100% on held probes; with eight documents this is a plumbing check, not evidence of retrieval quality—hard-negative evaluation is future work. Four physics tools (power flow, grid queries, contingency N-1, OPF) are validated 21/21 including 15/15 contingency convergences.

VI. EVALUATION PROTOCOL

Models. Two deployment tiers of the same prompt contract are compared: a lightweight API model (Gemini 3.5 Flash Lite, temperature 0, ≤ 512 output tokens, checkpointed rate-limit retry) and a small 4-bit-quantized open-weight model (Gemma 4 E4B-it [24]) served locally from a GGUF build via llama.cpp [25] (temperature 0, $\leq 1,024$ tokens; parsing reads content with fallback to its visible reasoning channel). *The API tier is a lightweight, latency-optimized model:* the comparison characterizes a representative local deployment against a budget API tier, not against frontier API models. A deterministic rule-based *oracle* (not a model) validates harness

plumbing at $N=600/\text{config}$; statistics on it test code, not intelligence.

Metrics. Diagnosis correctness is exact class match. *Tool selection is reported under two definitions.* (a) *Accepted-set:* the stated tool is required or strongly appropriate. This definition is degenerate—power flow lies in every scenario’s accepted set, so always answering power flow scores 100%—and we treat it as an upper bound only. (b) *Strict-specific:* the stated tool must be *required* for the scenario, and power flow counts only when it is the sole required tool. Bare `grid_query` answers are scored non-specific. Hallucination is response-level, any of six tags (H-NUM/TOP/EQP/PHY/TOOL/ACT), assigned by an automated rule-based judge; rates are judge agreement, not ground truth (Sec. VIII).

Power. At $n=140$ paired scenarios with the observed discordance rates, the minimum detectable difference (two-sided, $\alpha=0.05$, power 0.80) is E1: 2.8, E2: 2.0, E3: 3.9, E4: 2.8 pp. Null results in this paper therefore mean “no difference above the per-configuration MDE was detectable,” not “the models are equivalent.”

VII. RESULTS

A. A Label-Axis Failure Mode: E6 and E8

Table III shows near-ceiling diagnosis on every cause-labeled class and total collapse on two outcome-labeled classes: E6 (undervoltage) and E8 (thermal overload) are diagnosed correctly in 0 of 36 and 0 of 26 pooled observations. The failure is systematic, not stochastic: across all 144 failed E6 responses, 128 predict E3, 16 predict E4—and across all 104 failed E8 responses, 101 predict E9, 3 predict E3. The logged reasons are explicit: “The injected event is a single transmission line outage (line_6_13), which directly corresponds to class E3” (an E6 scenario); “the injected event description explicitly states a compound mechanism consisting of a transmission line outage and a 20% demand increase” (an E8 scenario).

The mechanism is a mixed-axis label taxonomy. E6/E8 are *outcome* classes—defined by the post-event state—but are *injected* through cause mechanisms (line outages that cause undervoltage; line-outage-plus-load-surge compounds that cause overloads). Both models classify on the cause axis and are therefore *causally correct but label-wrong* on every E6/E8 scenario. This is a property of the benchmark’s labeling scheme, not evidence about model capability: mixed-axis taxonomies silently reward or punish answer styles unrelated to the construct being measured. Correcting it requires either relabeling E6/E8 as cause events or instructing the taxonomy’s outcome primacy explicitly; both are queued for the next corpus revision, and the powered sweep will report both cause-axis and label-axis accuracy.

B. Cross-System Check: IEEE-39 (RQ7: Cross-System Transfer, Begun)

A seeded 140-scenario draw from the IEEE-39 corpus (excluding the 41 islanding-NaN scenarios) runs the same paired protocol. The API model diagnoses 107/140 scenarios in

TABLE III
E4 PER-CLASS, POOLED MODELS (RAW COUNTS; NO PERCENTAGES).

Class	Obs.	Diag.	Tool (strict)
E0	10	10/10	0/10
E1	26	26/26	0/26
E2	20	20/20	0/20
E3	28	28/28	28/28
E4	32	32/32	26/32
E5	38	38/38	0/38
E6	36	0/36	0/36
E7	22	22/22	0/22
E8	26	0/26	19/26
E9	42	40/42	26/42

Obs. = scenarios $\times$ models. Classes absent from the pilot sample: none.

TABLE IV
DIAGNOSIS, EXACT COUNTS WITH WILSON 95% CIs.

Cfg	API ($n=140$)	Local ($n=140$)
E1	109/140 [70, 84]	107/140 [69, 83]
E2	109/140 [70, 84]	108/140 [70, 83]
E3	109/140 [70, 84]	105/140 [67, 81]
E4	109/140 [70, 84]	107/140 [69, 83]

every configuration (E1–E4); the local model diagnoses 103–106/140 across configurations. Strict tool selection declines monotonically for the API model—50, 43, 38, and 27 of 140 across E1–E4—while the local model stays flat at 53–54/140. The decline extends the IEEE-14 pattern (53, 50, 36, and 35 of 140 there) but is steeper and does not plateau by E4, while the local model is flat on both corpora: accumulating context and tool manifests appears to progressively disrupt the API tier’s strict tool selection—flagged for the powered sweep. Transferring between corpora changes the network topology, load patterns, and label instances while the agent, prompt, and scoring pipeline remain untouched—the strongest test this pilot size permits.

C. RQ1/RQ4: Event Diagnosis and the Effect of Retrieval

Table IV reports exact counts with Wilson 95% CIs. The API model diagnoses 109/140 (77.9%) in every configuration; the local model 105–108/140 (75.0–77.1%). All diagnosis discordances are one-directional (2/1/4/2 across E1–E4: wherever the models disagree, the API model is right), and exact McNemar tests detect no significant difference (Table VI)—as expected at this power: the minimum detectable difference is E1: 2.8, E2: 2.0, E3: 3.9, E4: 2.8 pp. Configuration additions (retrieved procedures, tool manifests) changed not a single diagnosis for either model: the structured grid state in the prompt already carries the discriminating information, which answers RQ4 negatively *for diagnosis* at this corpus scale.

D. RQ2: Tool Selection and the Default-Answer Bias

Tool selection is where the deployment tiers differ most—in *style* before *accuracy*. Under the permissive accepted-set metric both models score 100%: the metric is degenerate, since always answering power flow achieves it by construction (random tool choice already scores 53%). The API model exercises exactly this strategy: 52% of its stated tools are power

TABLE V
TOOL SELECTION UNDER THE STRICT-SPECIFIC METRIC (DEFINED IN THE EVALUATION PROTOCOL), EXACT COUNTS WITH WILSON 95% CIs.

Cfg	API ($n=140$)	Local ($n=140$)
E1	53/140 [30, 46]	64/140 [38, 54]
E2	50/140 [28, 44]	63/140 [37, 53]
E3	36/140 [19, 34]	64/140 [38, 54]
E4	35/140 [19, 33]	64/140 [38, 54]

TABLE VI
EXPLORATORY PAIRED ANALYSIS (LOCAL – API DIAGNOSIS, PERCENTAGE POINTS). BOOTSTRAP 95% CI, 20K RESAMPLES. MARGINS WERE NOT PRE-REGISTERED; NO PASS/FAIL IS CLAIMED.

Cfg	Pairs	Diff (pp)	95% CI	McNemar p
E1	140	-1.4	[-3.6, 0.0]	0.50
E2	140	-0.7	[-2.1, 0.0]	1.00
E3	140	-2.9	[-5.7, -0.7]	0.12
E4	140	-1.4	[-3.6, 0.0]	0.50

flow. Under the strict-specific metric, the local model names a scenario-*required* tool in 63/140–64/140 of cases (flat across configurations) versus 35/140–53/140 for the API model. The conditioning behind these rates sharpens the default-answer-bias reading. When the API model states power flow (291 responses), that tool is actually required for the scenario in only 35% of cases—the rest are defaults to the universally accepted answer. The local model states power flow rarely (49 responses, 24% of them required); its dominant choices are contingency (327 responses, 78% required—contingency is required precisely on outage and overload scenarios) and equipment queries. The gap therefore reflects the API model’s default rate, not superior local tool knowledge; both models remain far from ceiling, and the binary metric cannot distinguish well-placed tools from lucky guesses.

E. RQ3: Tool-Call Execution Validity

The harness executes every stated tool on the scenario’s post-event network and scores the call. Execution is a property of the tool and scenario, not the model, so pairs are pooled across models and configurations: 518 stated-tool pairs, of

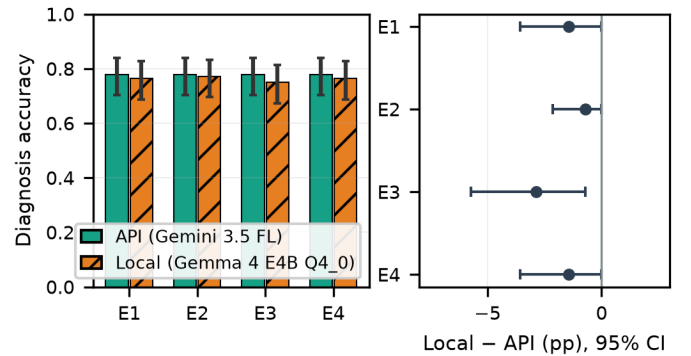


Fig. 3. (a) Diagnosis accuracy, local vs. API model, Wilson 95% CIs. (b) Paired diagnosis difference (Local – API) with bootstrap 95% CIs.

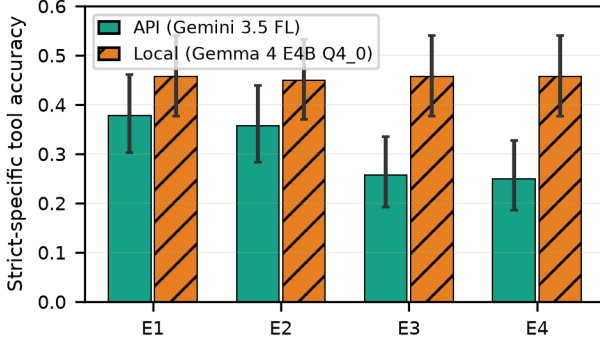


Fig. 4. Strict-specific tool-selection accuracy under the strict-specific metric (reference required set). Exact counts in Table V.

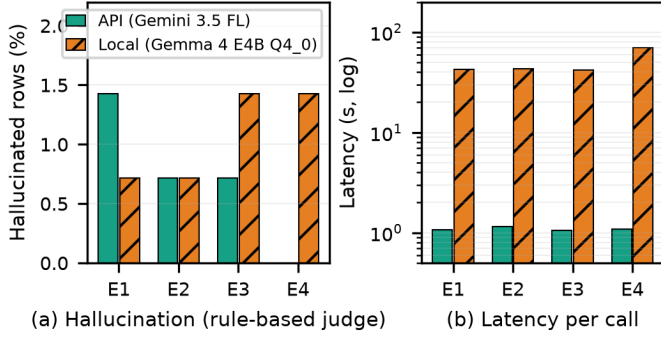


Fig. 5. Response-level any-tag hallucination (automated rule-based judge) and latency (log scale). Exact denominators in Table IV.

which 504 are executable (the remainder specified no tool) and every executable call succeeds—power flow converges on every reconstructed network, contingency executions converge and reproduce the scenario’s recorded overload, N-1 sweeps complete all line outages, and OPF solves. RQ3 is therefore answered affirmatively at the tool-name/execution level: every stated tool executes without error. Execution is, however, only a necessary condition for the valid tool-call construction RQ3 asks about: whether arguments are correct and the call targets the right contingency element is not checked. Argument-level validity (component identifiers, subcommands) requires an extended prompt schema and remains deferred to the powered sweep.

F. RQ5: Hallucination and the Cost of Local Inference

Any-tag hallucinated rows are rare for both models under the automated judge: 2/1/1/0 (E1–E4) (API) and 1/1/2/2 (E1–E4) (Local) across E1–E4. These rates mean “the rule-based judge flagged no row”; they are not expert-annotated ground truth. Latency is the deployment trade-off: local inference averages 42–70 s per call versus 1.06–1.16 s for the API round-trip—roughly a 45 $\times$ difference—while carrying zero marginal API cost, no rate-limit dependency, and no grid data leaving the premises; acceptable for always-on monitoring, but neither model’s latency profile suits closed-loop use (Sec. VIII).

Harness oracle and calibration (integration test, not evidence). On the seeded 600-scenario oracle run the E1→E4

TABLE VII
LOGGED E4 RESPONSE, SCENARIO IEEE14_SCN_002952 (API MODEL).

Field	Logged content
Answer	“{“event_class”: “E9”, “confidence”: 1.0, “tool”: “power_flow”, “reason”: “The injected description explicitly specifies a compound event involving both a transmission line outage and a load surge.” “
Judge	diag correct (E9); tool = power_flow $\in$ accepted set; no H-TOP/H-TOOL flags

ladder moves 55.7%→88.5% diagnosis (McNemar $p = 3.1 \times 10^{-32}$), validating that the harness can *detect* configuration differences; because the oracle is programmed to respond differently when RAG/tools are present, these statistics carry no evidence about real LLMs. Likewise, the oracle’s ECE numbers characterize its confidence simulator, not any model; at pilot N with near-saturated per-class correctness, model calibration is unidentifiable and will be reported on the powered sweep.

Qualitative trace. Table VII quotes the actual logged E4 response for scenario IEEE14_SCN_002952 ($V_{\min}$ 0.970 pu, $V_{\max}$ 1.060 pu, peak loading 1.4%, violations 0). The agent loop does not execute tool calls; the “tool” field is the model’s stated choice, whose execution validity is checked post hoc (Sec. VII-E).

VIII. DISCUSSION AND LIMITATIONS

Scope. The evidence covers *one model pair* (a 4-bit small local model; a lightweight API tier) on *one corpus*, one prompt template, temperature 0, single runs. Results do not speak to frontier API models or claim that local models match API models in general. The minimum detectable difference at this sample size is $\sim 3pp$ (Sec. VI); the powered 600-scenario sweep is the inferential step.

Label circularity. Disturbances, labels, and prompts derive from the same rule family: the evaluation measures *recovery of the synthetic labeling policy*, not open-ended operator reasoning—and the severity boundaries of Eq. (4) were themselves tuned on this corpus, though the strict-metric comparison is invariant across four boundary sets, two of them not derived from this corpus (Sec. IV). Required remedies, all future work: expert-reviewed labels (2×80 , $\kappa \geq 0.8$), multiple valid tool sequences, blind scoring, an evaluator independent of the label generator, and externally anchored severity boundaries.

Taxonomy design. The E6/E8 label-axis failure (Sec. VII-A) is a benchmark-design finding: outcome-labeled classes injected via cause mechanisms make cause-correct answers score zero. The next corpus revision separates the axes explicitly.

Physics fidelity. Thermal limits differ per system by construction, so overload magnitudes are not operationally comparable across systems. The corpus is static: no dynamics, protection, or time-sequential behavior. The severity index is illustrative with a quantified label-noise bound (Sec. IV);

the state estimator is a closed-form approximation pending replacement with pandapower’s iterative estimator; the 41 islanding-NaN IEEE-39 scenarios (identifiers shipped in `case39_nan_scenarios.csv`) remain unresolved and are excluded from downstream post-voltage use.

Scoring subjectivity. Hallucination tags come from a single automated rule-based judge (Sec. VII-F); tool scoring relies on the reference policy, and the strict metric narrows but does not eliminate policy-dependence.

Safety framing. This is an offline advisory prototype: no switching validation, protection coordination, authorization, or human-approval interface is implemented or claimed; closed-loop use is out of scope.

Reproducibility and generalization. All generation scripts, seeds, hashes, prompt templates, validation summaries, and the IEEE-39 NaN identifier list are in the repository; corpora regenerate deterministically; an archived (DOI) release is planned. Generalization: the IEEE-39 cross-system evaluation (Sec. VII) is performed for both deployment tiers; IEEE-118 transfer and a powered cross-system protocol remain future work, alongside the powered 600-scenario $\times$ 4-configuration sweep per model with per-class F1 under both label axes, ECE, multiplicity-corrected paired tests, and prompt/temperature sensitivity.

IX. CONCLUSION

GridPowerAgent couples a seeded, fully regenerable 16k/15k power-system scenario corpus with a grid-aware LLM agent that retrieves operating procedures and orchestrates validated physics tools, evaluated under a protocol where every number carries an exact denominator and every null result carries its statistical power. In the paired pilot, diagnosis was indistinguishable across deployment tiers, tool choice differed mainly in default-answer bias, and a systematic failure mode was traced to the benchmark’s own label taxonomy. The agent, the corpus, and the protocol—including their disclosed flaws—are the contribution; the powered evaluation they now enable is the next step.

AI Assistance Disclosure: Generative AI was used for drafting, coding assistance, and figure generation; all experimental decisions, execution, and analysis were performed by the authors, who reviewed and edited all AI-assisted content.

REFERENCES

- [1] S. Majumder, L. Dong, F. Doudi, Y. Cai, C. Tian, D. Kalathil, K. Ding, A. A. Thatte, N. Li, and L. Xie, “Exploring the capabilities and limitations of large language models in the electric energy sector,” *Joule*, vol. 8, pp. 1544–1549, 2024.
- [2] Y. Cheng, H. Zhao, X. Zhou, J. Zhao, Y. Cao, and C. Yang, “A large language model for advanced power dispatch,” *Scientific Reports*, vol. 15, p. 8925, 2025.
- [3] Y. Zhang, A. M. Saber, A. Youssef, and D. Kundur, “Grid-agent: An LLM-powered multi-agent system for power grid control,” *arXiv preprint arXiv:2508.05702*, 2025.
- [4] Y. Wen and X. Chen, “X-GridAgent: An LLM-powered agentic AI system for assisting power grid analysis,” *arXiv preprint arXiv:2512.20789*, 2025.
- [5] X. Zhou, H. Zhao, Y. Cheng, Y. Cao, G. Liang, G. Liu, W. Liu, Y. Xu, and J. Zhao, “ElecBench: A power dispatch evaluation benchmark for large language models,” *arXiv preprint arXiv:2407.05365*, 2024.
- [6] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 9459–9474.
- [7] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafraan, K. Narasimhan, and Y. Cao, “ReAct: Synergizing reasoning and acting in language models,” in *International Conference on Learning Representations (ICLR)*, 2023.
- [8] T. Schick, J. Dwivedi-Yu, R. Dessi, R. Raileanu, M. Lomeli, L. Zettlemoyer, N. Cancedda, and T. Scialom, “Toolformer: Language models can teach themselves to use tools,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023.
- [9] J. Achiam, S. Adler, S. Agarwal, L. Ahmad, I. Akkaya, F. L. Aleman *et al.*, “GPT-4 technical report,” *arXiv preprint arXiv:2303.08774*, 2023.
- [10] A. Abur and A. Gómez-Expósito, *Power System State Estimation: Theory and Implementation*. Marcel Dekker, 2004.
- [11] A. Monticelli, *State Estimation in Electric Power Systems: A Generalized Approach*. Springer, 1999.
- [12] A. J. Wood, B. F. Wollenberg, and G. B. Sheblé, *Power Generation, Operation, and Control*, 3rd ed. Wiley, 2014.
- [13] S. Frank, I. Steponavice, and S. Rebennack, “Optimal power flow: a bibliographic survey I,” *Energy Systems*, vol. 3, no. 3, pp. 221–289, 2012.
- [14] R. D. Zimmerman, C. E. Murillo-Sánchez, and R. J. Thomas, “MATPOWER: Steady-state operations, planning, and analysis tools for power systems research and education,” *IEEE Transactions on Power Systems*, vol. 26, no. 1, pp. 12–19, 2011.
- [15] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 2017, pp. 1321–1330.
- [16] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. J. Bang, A. Madotto, and P. Fung, “Survey of hallucination in natural language generation,” *ACM Computing Surveys*, vol. 55, no. 12, pp. 1–38, 2023.
- [17] A. Marot, B. Donnot, G. Dulac-Arnold, A. Kelly, A. O’Sullivan, J. Viebahn, M. Awad, I. Guyon, and B. Donon, “Learning to run a power network challenge for training topology controllers,” *Electric Power Systems Research*, vol. 189, p. 106635, 2020.
- [18] B. Donnot *et al.*, “Grid2Op: A testbed platform to model sequential decision making in power systems,” <https://github.com/rte-france/grid2op>, 2020.
- [19] L. Thurner, A. Scheidler, F. Schäfer, J.-H. Menke, J. Dollichon, F. Meier, S. Meinecke, and M. Braun, “pandapower—an open-source Python tool for convenient modeling, analysis, and optimization of electric power systems,” *IEEE Transactions on Power Systems*, vol. 33, no. 6, pp. 6510–6521, 2018.
- [20] J. Johnson, M. Douze, and H. Jégou, “Billion-scale similarity search with GPUs,” *IEEE Transactions on Big Data*, vol. 7, no. 3, pp. 535–547, 2021.
- [21] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence embeddings using siamese BERT-networks,” in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP-IJCNLP)*, 2019, pp. 3982–3992.
- [22] T. Dettmers and L. Zettlemoyer, “The case for 4-bit precision: k-bit inference scaling laws,” in *Proceedings of the 40th International Conference on Machine Learning (ICML)*, 2023, pp. 10775–10804.
- [23] E. Frantar, S. Ashkboos, T. Hoefler, and D. Alistarh, “GPTQ: Accurate post-training quantization for generative pre-trained transformers,” in *International Conference on Learning Representations (ICLR)*, 2023.
- [24] Gemma Team, Google DeepMind, “Gemma 4 technical report,” *arXiv preprint arXiv:2607.02770*, 2026.
- [25] G. Gerganov and contributors, “llama.cpp: LLM inference in C/C++,” <https://github.com/ggml-org/llama.cpp>, 2023, accessed 2026-08-31.